Problem 1

Question 1

output from running starter.py

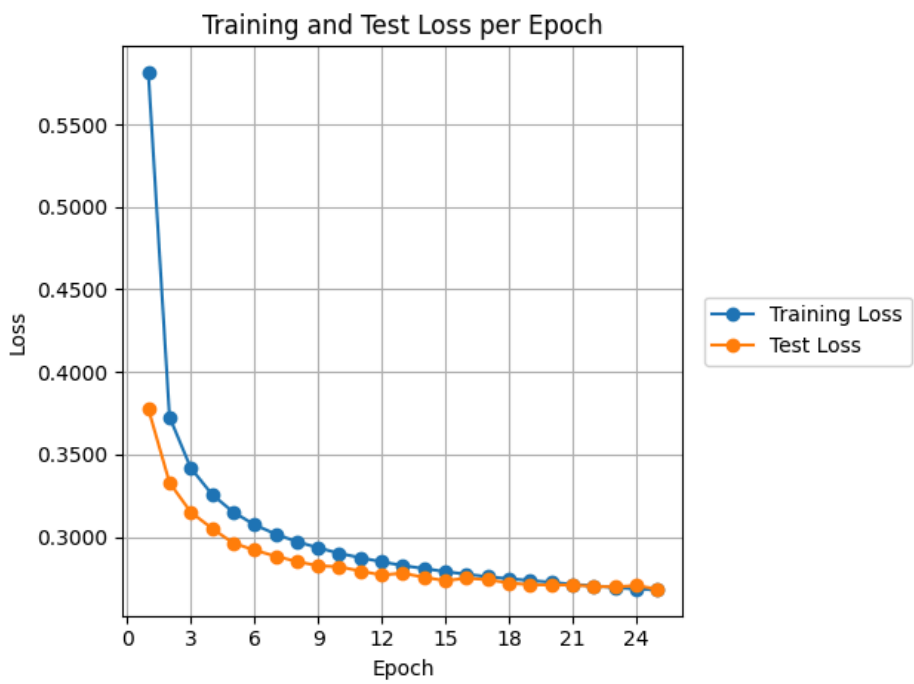
```
(HW1_virtualenv) login2.ls6(83)$ cat out.txt
Epoch: [1/25], Step: [100/468], Loss: 1.0163 Acc: 76.88%
Epoch: [1/25], Step: [200/468], Loss: 0.7770 Acc: 81.67%
Epoch: [1/25], Step: [300/468], Loss: 0.6714 Acc: 83.75%
Epoch: [1/25], Step: [400/468], Loss: 0.6103 Acc: 84.93%
Test accuracy: 90.12 % Test loss: 0.3777
Epoch: [2/25], Step: [100/468], Loss: 0.3956 Acc: 89.03%
Epoch: [2/25], Step: [200/468], Loss: 0.3799 Acc: 89.58%
Epoch: [2/25], Step: [300/468], Loss: 0.3768 Acc: 89.67%
Epoch: [2/25], Step: [400/468], Loss: 0.3768 Acc: 89.69%
Test accuracy: 91.01 % Test loss: 0.3330
Epoch: [3/25], Step: [100/468], Loss: 0.3372 Acc: 90.54%
Epoch: [3/25], Step: [200/468], Loss: 0.3437 Acc: 90.43%
Epoch: [3/25], Step: [300/468], Loss: 0.3427 Acc: 90.52%
Epoch: [3/25], Step: [400/468], Loss: 0.3427 Acc: 90.49%
Test accuracy: 91.21 % Test loss: 0.3151
Epoch: [4/25], Step: [100/468], Loss: 0.3387 Acc: 90.44%
Epoch: [4/25], Step: [200/468], Loss: 0.3327 Acc: 90.79%
Epoch: [4/25], Step: [300/468], Loss: 0.3320 Acc: 90.82%
Epoch: [4/25], Step: [400/468], Loss: 0.3281 Acc: 90.85%
Test accuracy: 91.52 % Test loss: 0.3053
Epoch: [5/25], Step: [100/468], Loss: 0.3144 Acc: 91.16%
Epoch: [5/25], Step: [200/468], Loss: 0.3170 Acc: 91.13%
Epoch: [5/25], Step: [300/468], Loss: 0.3186 Acc: 91.04%
Epoch: [5/25], Step: [400/468], Loss: 0.3148 Acc: 91.19%
Test accuracy: 91.78 % Test loss: 0.2964
Epoch: [6/25], Step: [100/468], Loss: 0.3156 Acc: 91.48%
Epoch: [6/25], Step: [200/468], Loss: 0.3071 Acc: 91.50%
Epoch: [6/25], Step: [300/468], Loss: 0.3076 Acc: 91.40%
Epoch: [6/25], Step: [400/468], Loss: 0.3082 Acc: 91.41%
Test accuracy: 91.84 % Test loss: 0.2921
Epoch: [7/25], Step: [100/468], Loss: 0.3035 Acc: 91.64%
Epoch: [7/25], Step: [200/468], Loss: 0.3037 Acc: 91.62%
Epoch: [7/25], Step: [300/468], Loss: 0.3035 Acc: 91.58%
Epoch: [7/25], Step: [400/468], Loss: 0.3006 Acc: 91.63%
Test accuracy: 91.90 % Test loss: 0.2884
Epoch: [8/25], Step: [100/468], Loss: 0.2903 Acc: 91.96%
```

```
Epoch: [8/25], Step: [200/468], Loss: 0.2908 Acc: 91.81%
Epoch: [8/25], Step: [300/468], Loss: 0.2963 Acc: 91.74%
Epoch: [8/25], Step: [400/468], Loss: 0.2967 Acc: 91.68%
Test accuracy: 91.94 % Test loss: 0.2853
Epoch: [9/25], Step: [100/468], Loss: 0.2952 Acc: 91.78%
Epoch: [9/25], Step: [200/468], Loss: 0.2964 Acc: 91.71%
Epoch: [9/25], Step: [300/468], Loss: 0.2966 Acc: 91.77%
Epoch: [9/25], Step: [400/468], Loss: 0.2933 Acc: 91.86%
Test accuracy: 92.15 % Test loss: 0.2826
Epoch: [10/25], Step: [100/468], Loss: 0.2981 Acc: 91.44%
Epoch: [10/25], Step: [200/468], Loss: 0.2940 Acc: 91.73%
Epoch: [10/25], Step: [300/468], Loss: 0.2922 Acc: 91.84%
Epoch: [10/25], Step: [400/468], Loss: 0.2913 Acc: 91.88%
Test accuracy: 91.97 % Test loss: 0.2821
Epoch: [11/25], Step: [100/468], Loss: 0.2911 Acc: 91.69%
Epoch: [11/25], Step: [200/468], Loss: 0.2878 Acc: 91.71%
Epoch: [11/25], Step: [300/468], Loss: 0.2873 Acc: 91.90%
Epoch: [11/25], Step: [400/468], Loss: 0.2877 Acc: 91.94%
Test accuracy: 92.19 % Test loss: 0.2794
Epoch: [12/25], Step: [100/468], Loss: 0.2975 Acc: 92.01%
Epoch: [12/25], Step: [200/468], Loss: 0.2908 Acc: 91.98%
Epoch: [12/25], Step: [300/468], Loss: 0.2868 Acc: 92.01%
Epoch: [12/25], Step: [400/468], Loss: 0.2849 Acc: 92.10%
Test accuracy: 92.07 % Test loss: 0.2770
Epoch: [13/25], Step: [100/468], Loss: 0.2772 Acc: 92.20%
Epoch: [13/25], Step: [200/468], Loss: 0.2775 Acc: 92.26%
Epoch: [13/25], Step: [300/468], Loss: 0.2789 Acc: 92.22%
Epoch: [13/25], Step: [400/468], Loss: 0.2831 Acc: 92.13%
Test accuracy: 92.21 % Test loss: 0.2782
Epoch: [14/25], Step: [100/468], Loss: 0.2892 Acc: 91.98%
Epoch: [14/25], Step: [200/468], Loss: 0.2867 Acc: 92.06%
Epoch: [14/25], Step: [300/468], Loss: 0.2820 Acc: 92.19%
Epoch: [14/25], Step: [400/468], Loss: 0.2822 Acc: 92.12%
Test accuracy: 92.23 % Test loss: 0.2757
Epoch: [15/25], Step: [100/468], Loss: 0.2873 Acc: 91.88%
Epoch: [15/25], Step: [200/468], Loss: 0.2836 Acc: 92.16%
Epoch: [15/25], Step: [300/468], Loss: 0.2818 Acc: 92.21%
Epoch: [15/25], Step: [400/468], Loss: 0.2792 Acc: 92.26%
Test accuracy: 92.34 % Test loss: 0.2737
Epoch: [16/25], Step: [100/468], Loss: 0.2931 Acc: 91.74%
Epoch: [16/25], Step: [200/468], Loss: 0.2842 Acc: 92.00%
Epoch: [16/25], Step: [300/468], Loss: 0.2800 Acc: 92.21%
Epoch: [16/25], Step: [400/468], Loss: 0.2789 Acc: 92.27%
Test accuracy: 92.17 % Test loss: 0.2751
```

```
Epoch: [17/25], Step: [100/468], Loss: 0.2784 Acc: 92.17%
Epoch: [17/25], Step: [200/468], Loss: 0.2720 Acc: 92.43%
Epoch: [17/25], Step: [300/468], Loss: 0.2744 Acc: 92.36%
Epoch: [17/25], Step: [400/468], Loss: 0.2759 Acc: 92.32%
Test accuracy: 92.35 % Test loss: 0.2743
Epoch: [18/25], Step: [100/468], Loss: 0.2632 Acc: 92.89%
Epoch: [18/25], Step: [200/468], Loss: 0.2726 Acc: 92.48%
Epoch: [18/25], Step: [300/468], Loss: 0.2757 Acc: 92.36%
Epoch: [18/25], Step: [400/468], Loss: 0.2749 Acc: 92.41%
Test accuracy: 92.27 % Test loss: 0.2722
Epoch: [19/25], Step: [100/468], Loss: 0.2745 Acc: 92.33%
Epoch: [19/25], Step: [200/468], Loss: 0.2705 Acc: 92.55%
Epoch: [19/25], Step: [300/468], Loss: 0.2701 Acc: 92.46%
Epoch: [19/25], Step: [400/468], Loss: 0.2748 Acc: 92.32%
Test accuracy: 92.28 % Test loss: 0.2712
Epoch: [20/25], Step: [100/468], Loss: 0.2650 Acc: 92.66%
Epoch: [20/25], Step: [200/468], Loss: 0.2628 Acc: 92.66%
Epoch: [20/25], Step: [300/468], Loss: 0.2693 Acc: 92.53%
Epoch: [20/25], Step: [400/468], Loss: 0.2706 Acc: 92.47%
Test accuracy: 92.25 % Test loss: 0.2710
Epoch: [21/25], Step: [100/468], Loss: 0.2675 Acc: 92.56%
Epoch: [21/25], Step: [200/468], Loss: 0.2725 Acc: 92.35%
Epoch: [21/25], Step: [300/468], Loss: 0.2747 Acc: 92.33%
Epoch: [21/25], Step: [400/468], Loss: 0.2722 Acc: 92.39%
Test accuracy: 92.16 % Test loss: 0.2710
Epoch: [22/25], Step: [100/468], Loss: 0.2764 Acc: 92.63%
Epoch: [22/25], Step: [200/468], Loss: 0.2729 Acc: 92.60%
Epoch: [22/25], Step: [300/468], Loss: 0.2718 Acc: 92.55%
Epoch: [22/25], Step: [400/468], Loss: 0.2705 Acc: 92.54%
Test accuracy: 92.37 % Test loss: 0.2702
Epoch: [23/25], Step: [100/468], Loss: 0.2640 Acc: 92.62%
Epoch: [23/25], Step: [200/468], Loss: 0.2646 Acc: 92.61%
Epoch: [23/25], Step: [300/468], Loss: 0.2674 Acc: 92.56%
Epoch: [23/25], Step: [400/468], Loss: 0.2664 Acc: 92.54%
Test accuracy: 92.29 % Test loss: 0.2699
Epoch: [24/25], Step: [100/468], Loss: 0.2715 Acc: 92.48%
Epoch: [24/25], Step: [200/468], Loss: 0.2768 Acc: 92.37%
Epoch: [24/25], Step: [300/468], Loss: 0.2678 Acc: 92.65%
Epoch: [24/25], Step: [400/468], Loss: 0.2681 Acc: 92.60%
Test accuracy: 92.28 % Test loss: 0.2706
Epoch: [25/25], Step: [100/468], Loss: 0.2686 Acc: 92.57%
Epoch: [25/25], Step: [200/468], Loss: 0.2692 Acc: 92.55%
Epoch: [25/25], Step: [300/468], Loss: 0.2708 Acc: 92.54%
Epoch: [25/25], Step: [400/468], Loss: 0.2700 Acc: 92.56%
```

Test accuracy: 92.35 % Test loss: 0.2686

Question 2

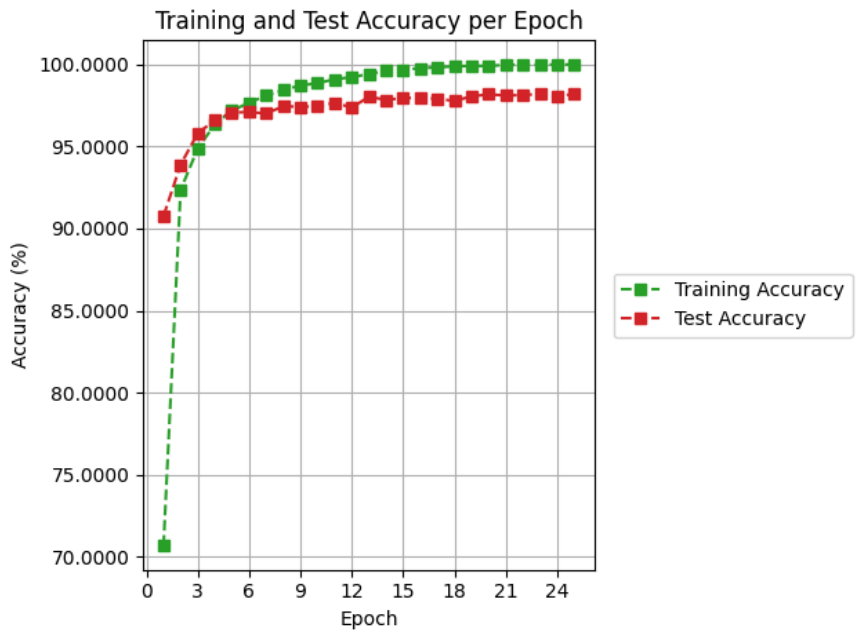


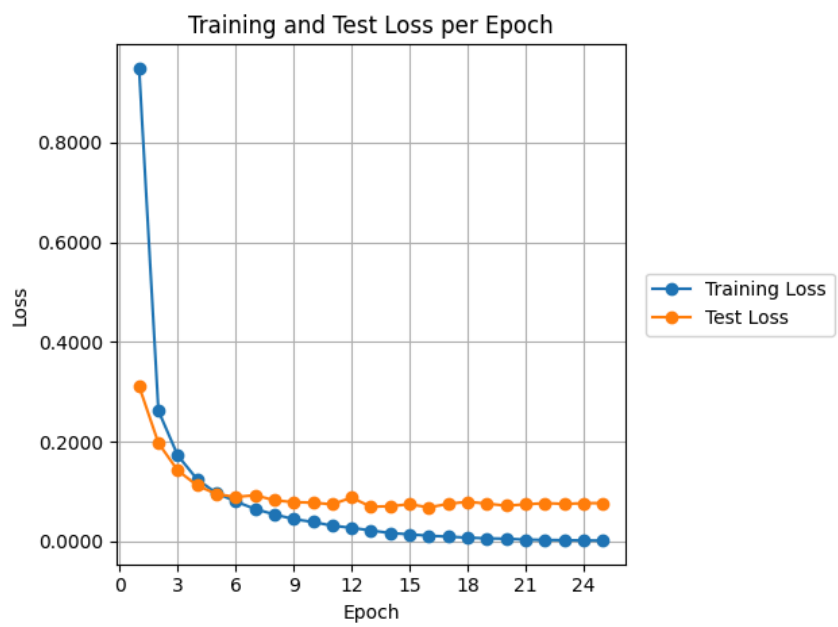
Question 3

Training accuracy [%]	Testing accuracy [%]	Total time for training [s]	Total time for inference [s]	Average time for inference per image [ms]	GPU memory during training [MB]
92.565	92.35	202.1325	1.1750	0.1175	537

Problem 2

Question 1





This model definitely overfits- training accuracy approaches 100 percent, while testing accuracy levels out around 98 percent. Similarly, the training loss approaches 0, while testing loss stops improving.

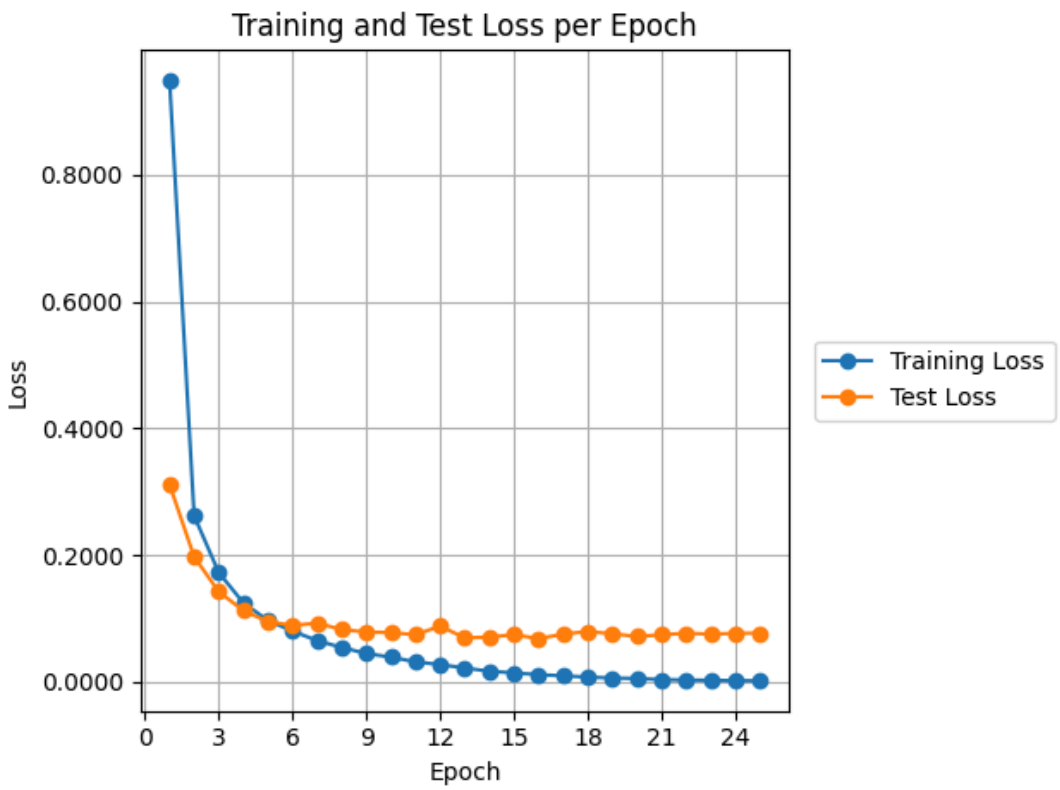
Question 2

Adjusted config.slurm to run for all p values in one run

```
python -u $WORK/HW1_files/simpleFC.py --batch_size=128 --epochs=25 --lr=0.01 --p=0.0 >
$WORK/HW1_files/out.txt
python -u $WORK/HW1_files/simpleFC.py --batch_size=128 --epochs=25 --lr=0.01 --p=0.2 >
$WORK/HW1_files/out.txt
python -u $WORK/HW1_files/simpleFC.py --batch_size=128 --epochs=25 --lr=0.01 --p=0.5 >
$WORK/HW1_files/out.txt
python -u $WORK/HW1_files/simpleFC.py --batch_size=128 --epochs=25 --lr=0.01 --p=0.8 >
$WORK/HW1_files/out.txt
```

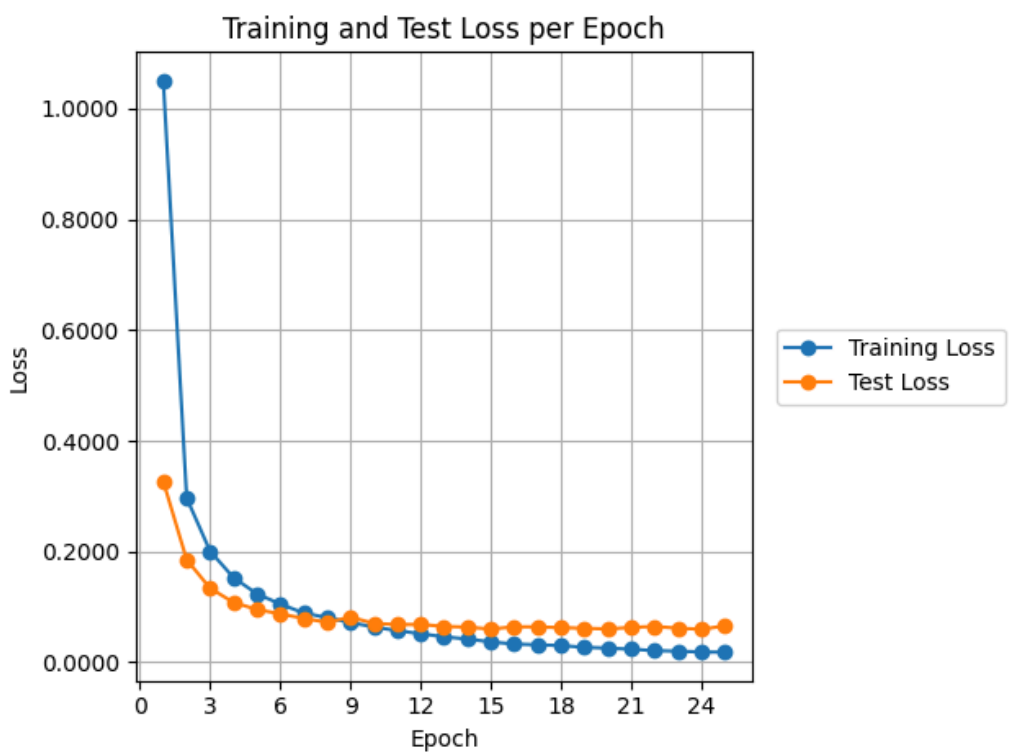
Plots for 0.0

Training and Test Accuracy per Epoch
Accuracy (%)
100.0000
95.0000
90.0000
85.0000
80.0000
75.0000
70.0000
0
3
6
9
12
15
18
21
24
Epoch
Training Accuracy
Test Accuracy

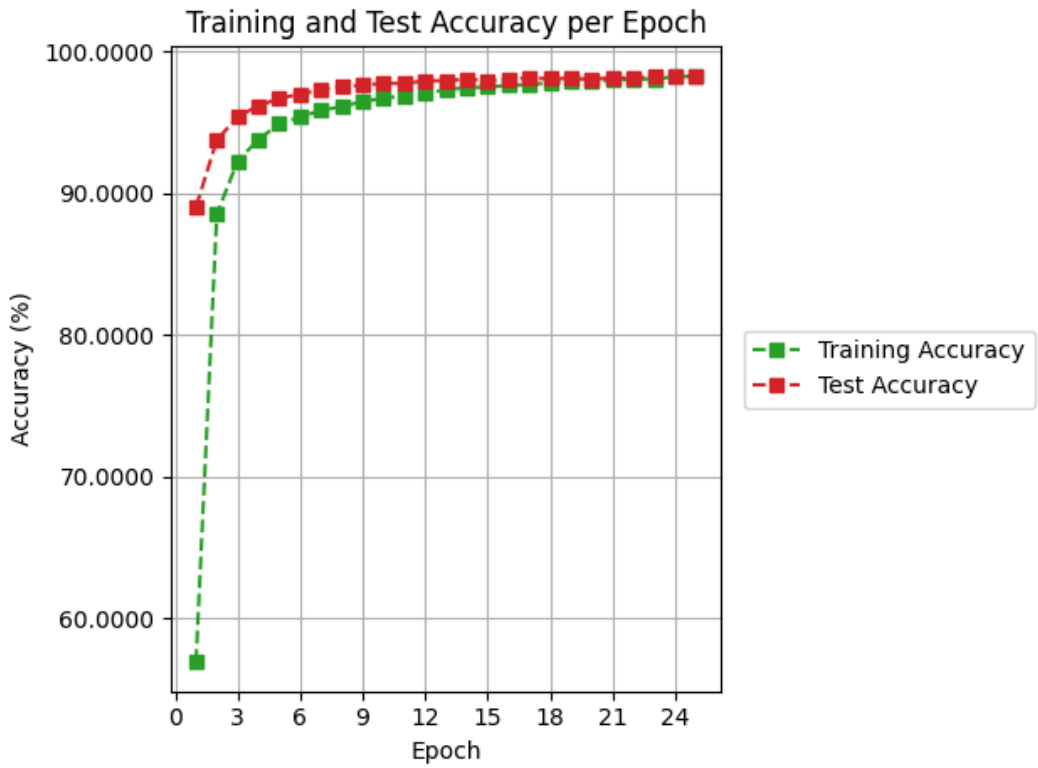

Training and Test Loss per Epoch
Loss
0.8000
0.6000
0.4000
0.2000
0.0000
0
3
6
9
12
15
18
21
24
Epoch
Training Loss
Test Loss

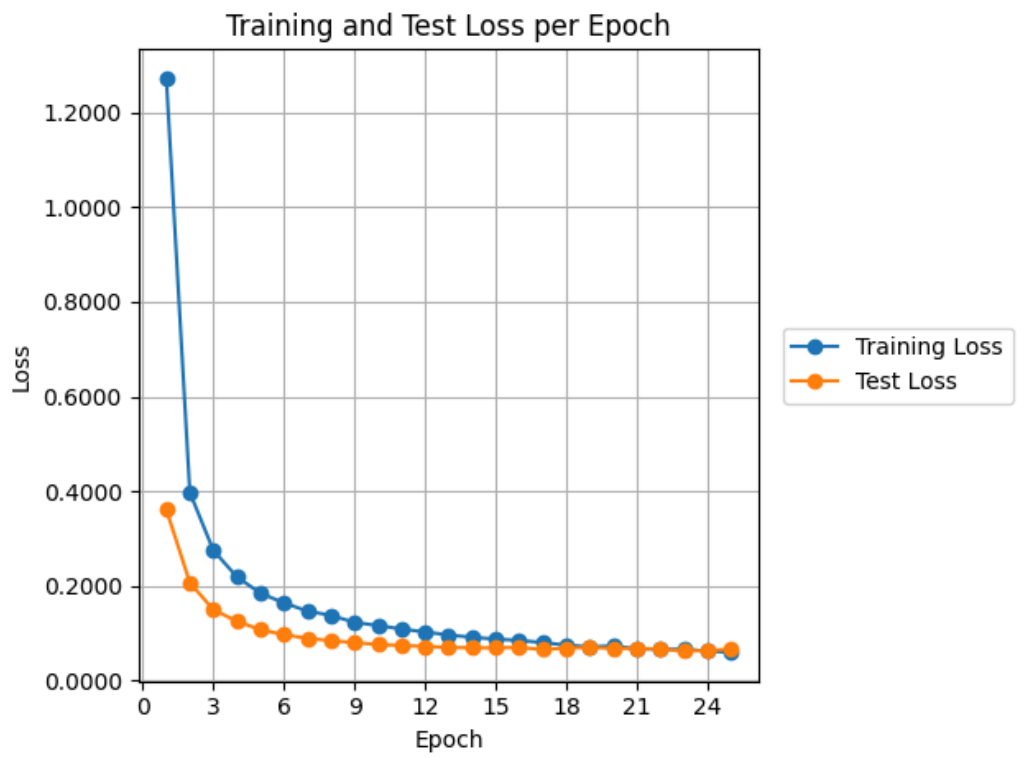
Plots for 0.2

Training and Test Accuracy per Epoch
100.0000
95.0000
90.0000
85.0000
80.0000
75.0000
70.0000
65.0000
Accuracy (%)
0
3
6
9
12
15
18
21
24
Epoch
Training Accuracy
Test Accuracy

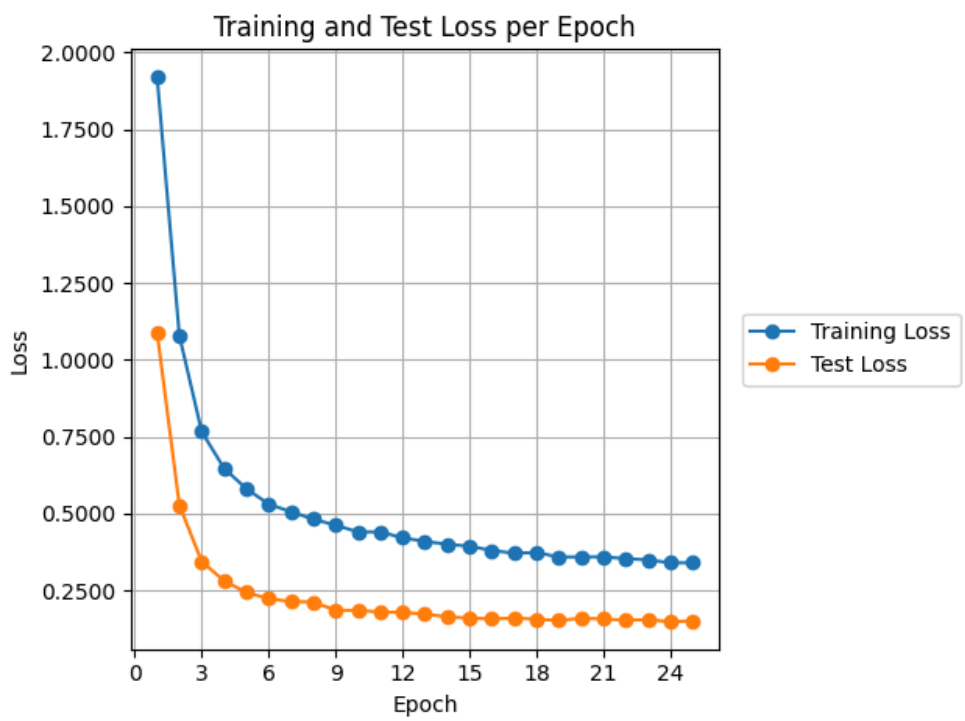

Training and Test Loss per Epoch
1.0000
0.8000
0.6000
0.4000
0.2000
0.0000
Loss
0
3
6
9
12
15
18
21
24
Epoch
Training Loss
Test Loss

Plots for 0.5


Training and Test Accuracy per Epoch
100.0000
90.0000
80.0000
70.0000
60.0000
Accuracy (%)
0
3
6
9
12
15
18
21
24
Epoch
Training Accuracy
Test Accuracy


Training and Test Loss per Epoch
1.2000
1.0000
0.8000
0.6000
0.4000
0.2000
0.0000
Loss
0
3
6
9
12
15
18
21
24
Epoch
Training Loss
Test Loss

Plots for 0.8



0.5 gave the best results. While 0.0, 0.2, and 0.5 all had similarly high test accuracies (~98 percent), the training accuracy matched the testing accuracy most closely for dropout rate 0.5.

The training accuracy in the other values pretty clearly surpasses the testing accuracy. On the other hand, 0.8 was not comparable in performance to the other three values (96 percent).

Question 3

Dropout	Training accuracy [%]	Testing accuracy [%]	Total time for training [s]	First epoch when the model reaches 96% training accuracy
0.5	98.2417	98.2800	233.1333	8
0.5 + norm	98.5900	98.4300	336.4420	6

Most metrics are improved when using normalization, including accuracy and first epoch to reach 96 percent accuracy. However, the training time increased substantially.

Problem 3

Question 1

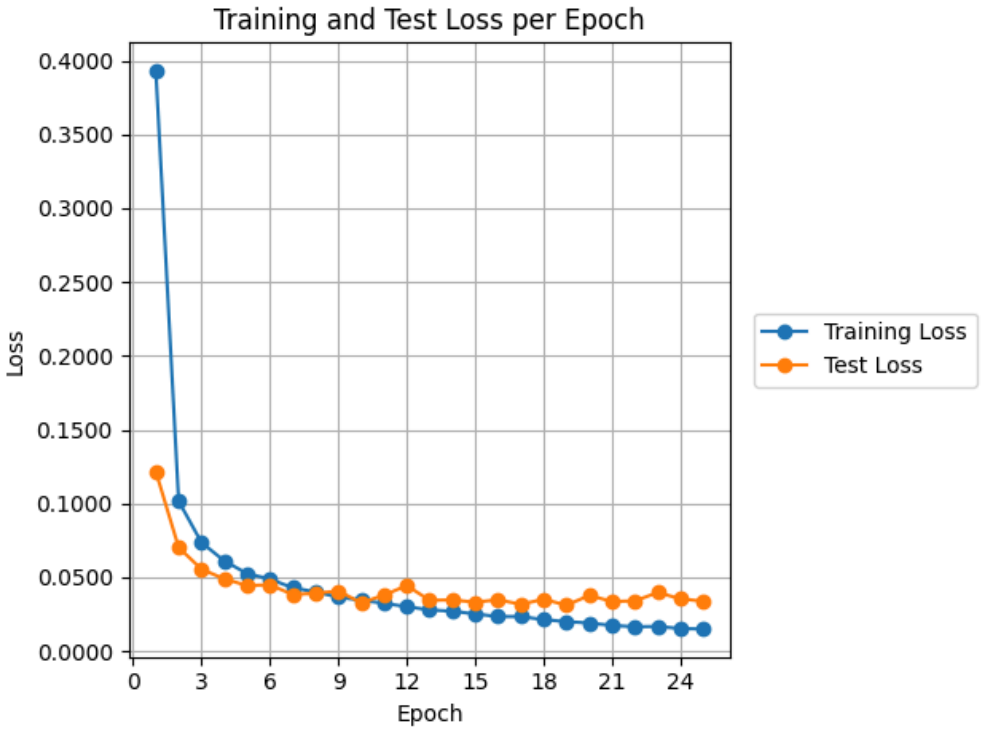
Model name	MACs	FLOPs	# parameters	**torchsummary** (total) size [KB]	Saved model size [KB]
SimpleCNN	3,869,824	7,739,648	50,186	196.04	201.36

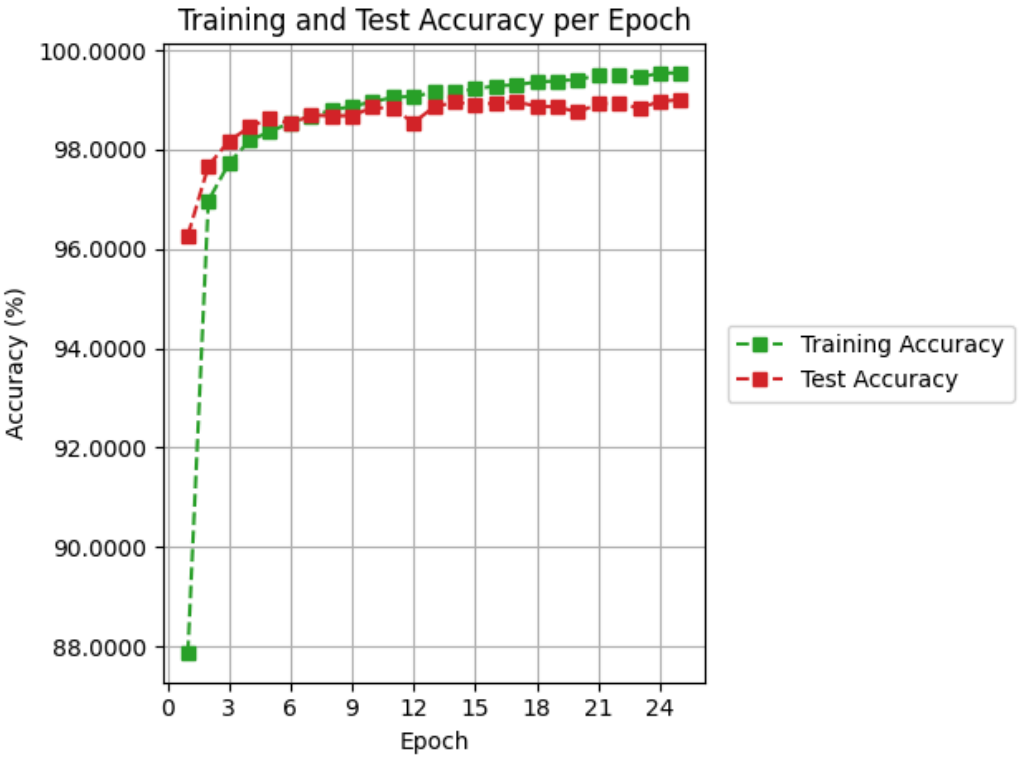
Torchsummary is smaller than the saved model size because it contains just parameters whereas the saved model contains the overhead of a pickled object.

Question 2

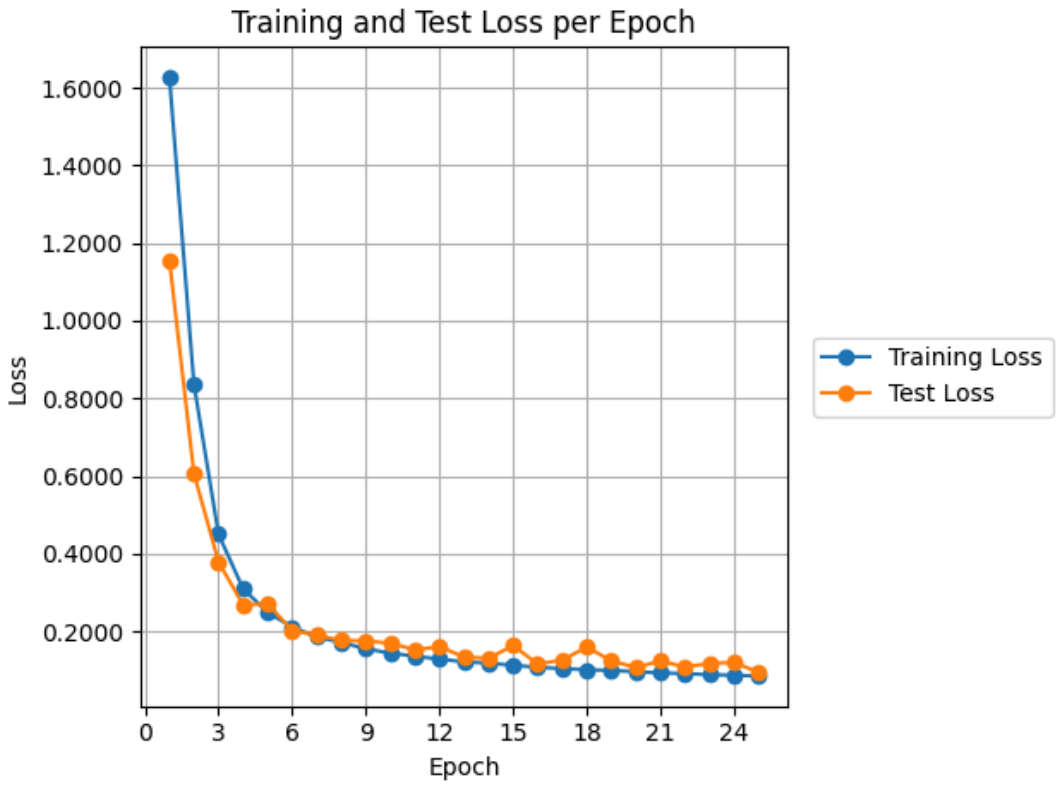
Model name	MACs	FLOPs	# parameters	**torchsummary** (total) size [KB]	Saved model size [KB]
SimpleCNN	3,869,824	7,739,648	50,186	196.04	201.36
myCNN	1,093,248	2,186,496	5,226	26.39	28.8

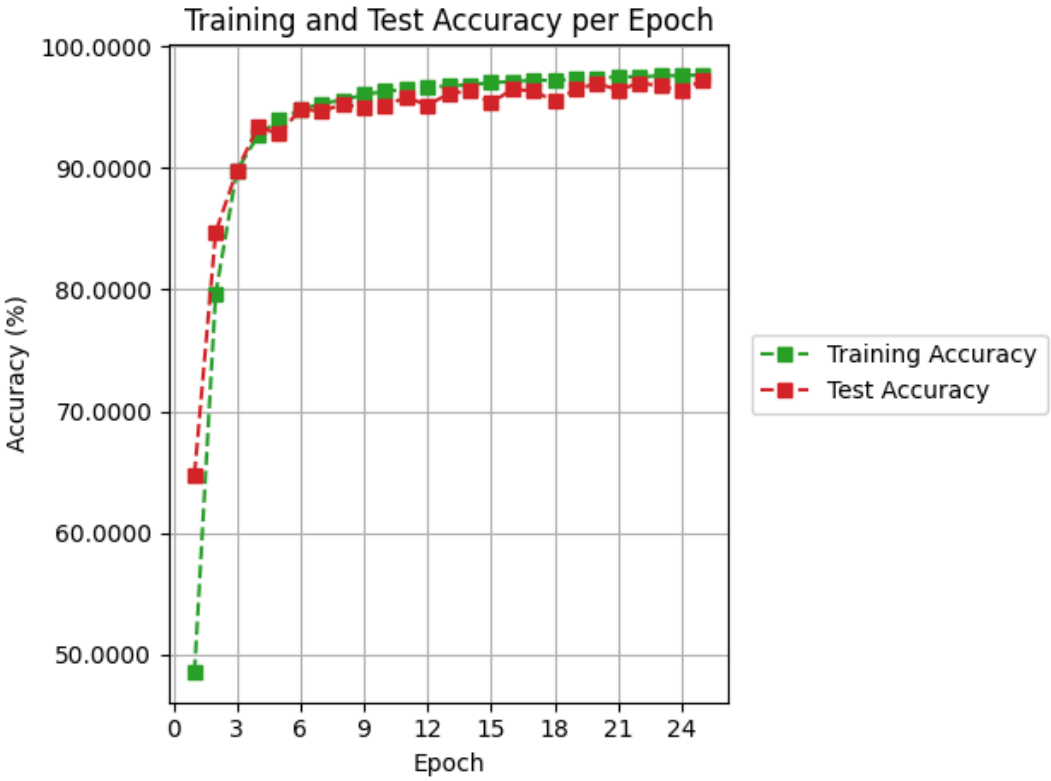
SimpleCNN Plots





MyCNN Plots


Training and Test Loss per Epoch
Loss
Epoch
1.6000
1.4000
1.2000
1.0000
0.8000
0.6000
0.4000
0.2000
0
3
6
9
12
15
18
21
24
Training Loss
Test Loss


Training and Test Accuracy per Epoch
Accuracy (%)
Epoch
100.0000
90.0000
80.0000
70.0000
60.0000
50.0000
0
3
6
9
12
15
18
21
24
Training Accuracy
Test Accuracy